



Society of Petroleum Engineers

SPE-226854-MS

Case Study: Controlling Injection in Horizontal Waterflood Wells Utilizing a Novel Inflow Control Devices: Challenge, Application, and Lessons Learned

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

This paper examines the implementation of Inflow Control Devices (ICDs) that have been used in horizontal water injection wells to balance outflow injected fluid performance along the lateral which Improves sweep efficiency and reduces the risk of early injected fluid breakthrough in producer wells. As erosion and plugging issues may accrue when fluids injected with high velocity and pressure drop across standard nozzle Inflow Control, a novel design featuring enhanced flow areas, the novel ICD, that provides a higher flow area which allows gradual pressure drop across ICD with much lower fluid velocity reducing the erosion/plugging risk.

The novel design of ICD Cassette utilizes advanced geometry to achieve desired flow restrictions that are equivalent to the conventional nozzle-based ICD. The novel ICD Cassette has four times the open flow area compared with equivalent nozzle-based ICDs, which leads to lower fluids velocities through the device, reducing erosion, and making the design less susceptible to plugging. In addition, the gradual pressure change through the novel ICD Cassette helps to reduce scaling tendency in the device, and therefore minimizing the number of cleanout intervention operations required throughout the well lifecycle. Design optimization for ICD completion involved compartmentalization using open hole logs for two wells considering targeted injection rates. Each injection compartment was equipped with adjustable flow control devices and isolated with packers to ensure injected fluid goes into desired zones. The ICD joint was equipped with sliding sleeves to enable operators to modify injection profiles or isolate compartments as needed, and by that adding flexibility to operations.

As the operation commenced in heavy oil reservoir with a high variety of permeability, the effectiveness of the novel ICD in achieving balanced injection distribution across multiple compartments was demonstrated. With the challenge of uncertainty in permeability profile and fluid saturation of would affect injection performance, utilizing equal choking strength cassettes allowed for decent distribution across injection compartments. Operator can modify compartment injection flow area since novel ICD joints are equipped with sliding sleeve or future complete shutoff of any intervals that prematurely breakthrough to producer wells. Injectivity test was performed after each sleeve was opened to confirm flow and to set a baseline to identify any future plugging of the flow control devices. During producing

stage, unexpected variations in oil viscosity, measuring between 400 to 1,000 cP, and along with varying oil saturations profiles, quality, caused inaccurate model pressure predictions. Despite these challenges, injectivity confirmed the operational integrity of the devices, allowing for compartmental specific flow adjustments and if necessary, isolating the underperforming zones.

Lessons learned from this deployment, for future wells installations and introducing a new design philosophy, Include the need for enhanced reservoir characterization using advanced logging techniques to refine the permeability profiles. This can also be performed by collecting samples to ensure accurate fluid properties and saturation. In future designs, using variable compartment injection flow area instead of uniform injection flow area distribution especially with enhancing used data for design optimization would ensure better alignment between operational outcomes and design expectations.

The paper is showcasing the advancements in water injection technology, by implementing the novel ICD design that mitigates the challenges erosion and plugging while offering improved scalability that is considered as a step changer in operational efficiency and reservoir management. This approach supports more sustainable waterflood operations by reducing maintenance and intervention needs while providing beneficial insights for optimizing future installations.

Introduction

There are typically two mechanisms to recover oil: conventional recovery and enhanced recovery Fig. (1). The conventional recovery mechanism has a primary and secondary stages, where the primary recovery relies on natural reservoir energy (such as solution gas drive, gas cap expansion, or aquifer influx), the resulting recovery factor often ranges from 5% to 40% of the original oil in place. Once this natural drive depletes, secondary recovery stages like waterflooding become essential to maintain reservoir pressure and pushing the remaining oil inside the reservoir which provide additional aid to increase oil recovery factor.

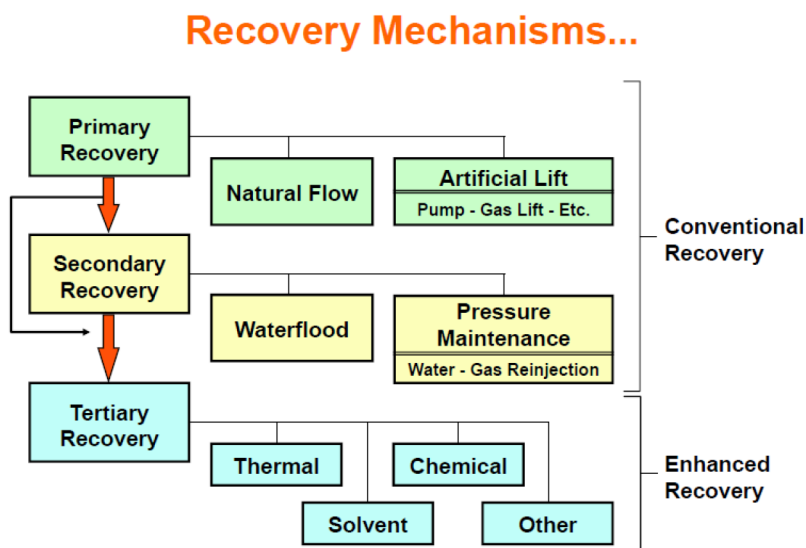


Figure 1—Different Reservoir Recover Mechanisms – Waterflood as Secondary Recovery (Abubaker, 2015)

Waterflooding, water injection to displace oil from pore spaces while sustain reservoir pressure, remains one of the most adopted techniques as a secondary stage. Its effectiveness is highly dependent on the rock and fluid properties like:

- **Oil properties** – viscosity (thickness) and density (weight) of oil
- **Water properties** – compatibility of injection water with existing water in formation

- **Rock properties** – permeability, porosity, heterogeneity (difference)

Operational challenges associated with waterflooding include uncontrolled injection, early water breakthrough, and inefficient sweep, that could affect efficiency of recovery optimization and therefore increasing operational cost. Figure (2) shows two injectors performance. Well-A had completed with nozzle-based inflow control devices (ICDs) downhole completion while Fig. (3) shows Well-B performance using the novel ICD which has higher flow area to overcome the challenge of high injection pressure and performance fluctuation which shown in Well-A Fig. (2). Water injection KPIs are used to evaluate wells injection performance such as Injection Index and Standard deviation.

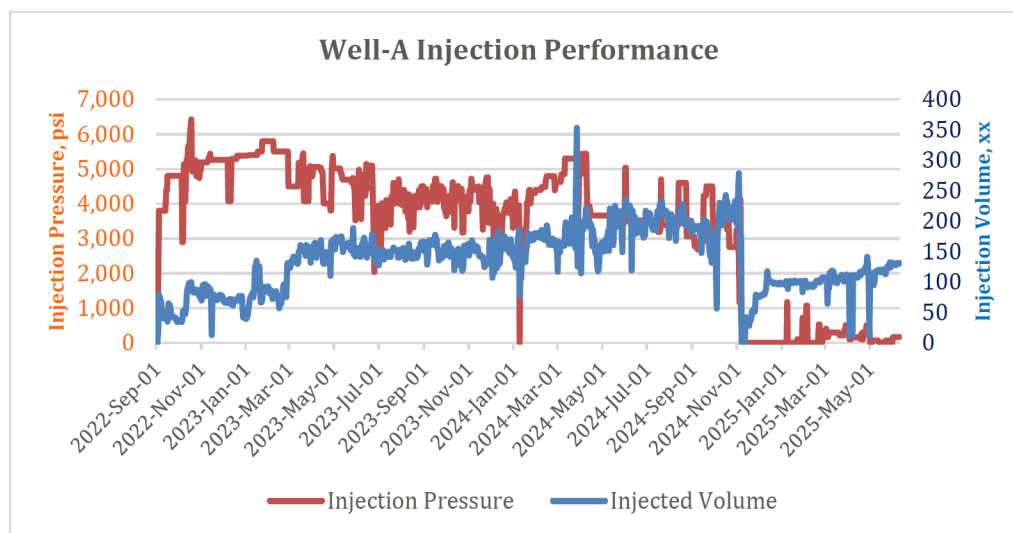


Figure 2—Well-A injection Performance – Nozzle-based downhole completion

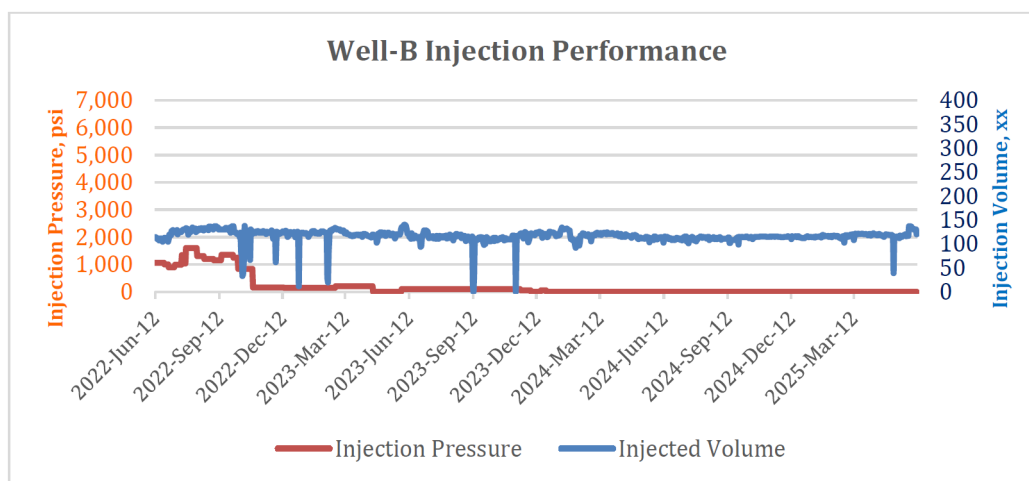


Figure 3—Well-B injection Performance – Novel ICD downhole completion

Injection Index tells you how efficiently a well accepts injected fluid; it calculated by dividing injected volume or rate per Injection pressure. The higher index the more volume per unit of pressure means Better injectivity while Lower injection index means requires more pressure for the same volume due to poorer injectivity or plugging

Standard deviation (e.g., of pressure or index) shows variability over time. Low standard deviation means Stable / predictable performance while High standard deviation means Fluctuations, possibly due to:

- Pressure surges
- Unstable formation
- Operational inconsistencies

Tables 1 and 2 compare injection parameters for both wells.

Table 1—Well-A; Water Injection KPI parameters

Well	Well-A			
Days	1014			
	<i>Min</i>	<i>Avg</i>	<i>Max</i>	<i>Total</i>
<i>Injected Volume, XX</i>	11.0	136.0	353.2	137885.4
<i>Injection Pressure; psi</i>	15.0	3807.5	6423.0	
<i>Injection Index, xx/day/psi</i>	0.0025	0.14	7.84	
<i>Injection Standard Deviation, psi/d</i>	6.9657	1432.09	1955.83	

Table 2—Well-B; Water Injection KPI parameters

Well	Well-B			
Days	1095			
	<i>Min</i>	<i>Avg</i>	<i>Max</i>	<i>Total</i>
<i>Injected Volume, XX</i>	11.7	117.1	140.2	128227.4
<i>Injection Pressure; psi</i>	50	414	1600	
<i>Injection Index, xx/day/psi</i>	0.0394	0.77	2.50	
<i>Injection Standard Deviation, psi/d</i>	2.1637	99.03	386.42	

Calculated KPI parameters shows that Well-B has average Injection index 5.5 higher than Well-A also Well-B shows more stable injection performance with injection standard deviation value less 15 times Well-A. These KPI parameters could be due to many reasons such as Plugging; Pressure buildup; Formation compaction and or different oil saturation around well bore hole for that reason operator looking for the new novel ICD type which has more flow area compared to nozzle based and installed in Well-B which shows much better performance compared to Well-A

These operational challenges are usually highlighted in bullhead injection scenarios, where water is injected without zonal control. As shown in Fig. 4, bullhead injection often results in uncontrolled fluid distribution, leading to overflooded zones, early water breakthrough, and ultimately inefficient reservoir sweep. The inability to manage zonal injection can also increase intervention costs, especially when attempting to shut off high water-cut zones later in the well life.

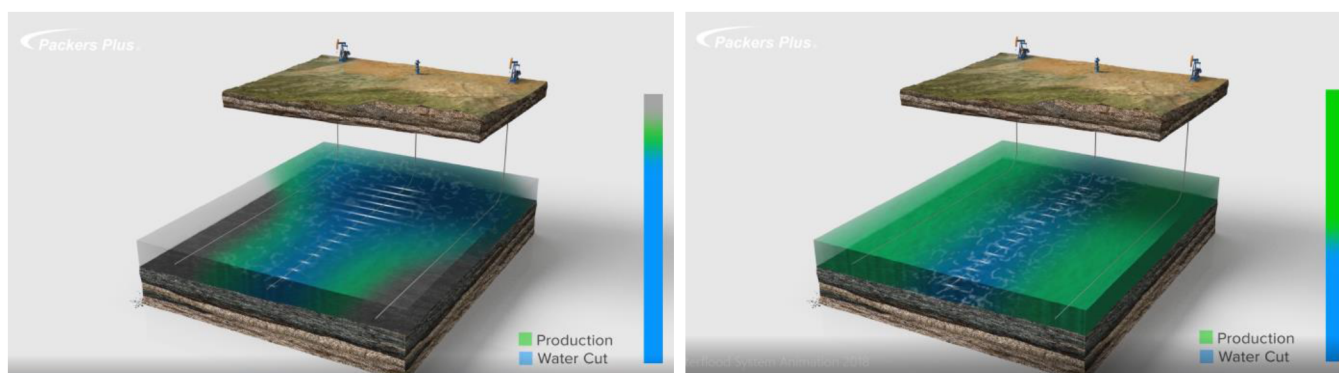


Figure 4—Bullhead Injection Simulation (Left) and Flow Control Device Injection Simulation (Right)

In comparison, injection systems equipped with Flow Control Devices (FCDs) offer a proactive solution. By distributing injected fluids more uniformly across multiple reservoir compartments, FCDs help mitigate these common issues. They contribute to:-

- More efficient sweep
- Delayed breakthrough at producer wells, and
- Higher ultimate recovery (EUR)

The novel ICD features advantages that for better gradual pressure drop and reduced flow velocity while having a bigger cross section flow area compared with equivalent nozzle size (same fluid performance). In addition, this would lower the risk of erosion, plugging, and improves scale resistance due to smoother pressure transition. The design also enables zonal flexibility through sliding sleeves to allow operators to selectively shut off the compartments in response to field performance.

Novel ICD Technology for Injector Wells

Novel Flow Control Devices is used to balance influx / outflux for both producer and injector wells via distribute flow along the liner into compartments. Each ICD joint comes with an on-off sleeve to shut-off compartments that may no longer need production / injection Fig. (5). The sleeve also can act as an isolation device to set liner hanger and hydromechanical packers. novel ICD can be opened / closed via a wireline or coiled tubing intervention using standard shifting tool.

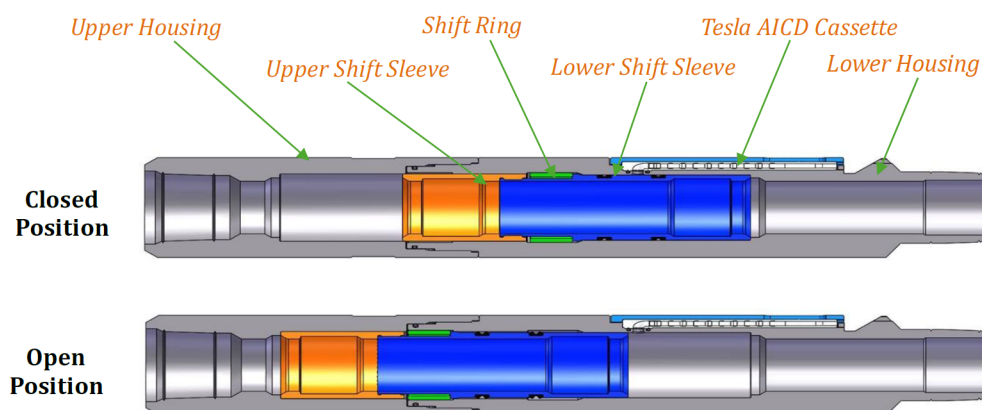


Figure 5—Novel ICD Joint Components for Injection applications

In injector wells, water is injected down the tubing string from surface; the Injection novel ICD downhole completion works to distribute the fluid flow along the well to gain the injection conformance desired. A variable pressure drop occurs across the ICD cassettes in each compartment that limits the amount of fluid exiting each compartment. The cassettes are modular, with the same outer dimensions. This allows interchangeability of different cassettes in the same chassis design. Different cassettes have different strengths that modeling will aid in placement. The Injection cassettes are assembled on the novel ICD joints based on open hole logs and design optimization which recommend the optimum number of cassettes per joint and optimum joints per compartment based on well target injection rate and operation condition. It is typically run in conjunction with annular isolation packers. the main objectives using novel ICD downhole completion are:-

- Balanced outflow conformance along the lateral to avoid early injected water breakthrough in produced wells.
- Pressure and flow rates can be custom designed based on open hole logs
- Reduce Nozzle plugging risk – Novel ICD Cassette has a higher open flow area compared to a nozzle.
- Down hole Flow Control completion should be high erosion resistance

The operator converts down hole flow control device from Nozzle to ICD cassette because it offers a large cross-sectional flow area compared to a nozzle. On average the ICD Profile supplies four times nozzle cross sectional flow area for the same pressure drop and thus reduces fluid velocity which has benefits in erosion and plugging resistance. There are currently six different cassette sizes and four places in joint to place ICD cassettes which allows flexibility in design optimization with more than one hundred possible joint flow area depend on cassettes number / size per joint. To achieve one of main operator completion objectives; cassette could be 3-D printing from different materials – Cobalt Chrome Moly, 17-4 PH Stainless, Inconel 718 to meet client erosion resistance.

The Novel ICD profile depends on Tesla concept which has been invented by Nikola Tesla in 1920, it has a higher choking capacity in one direction versus the other and this physical characteristic is called Diodicity. The choking characteristics of the profile results from fluid collisions taking place within the device. Fluid streams are split and then are jetted back upon themselves as the streams move through the cassette. The splitting and merging of the streams are what causes the pressure drop, rather than the fluid viscosity. Variable testing for different fluid viscosities proves that ICD profile is viscosity independent within the range of testing envelope Fig. (6)

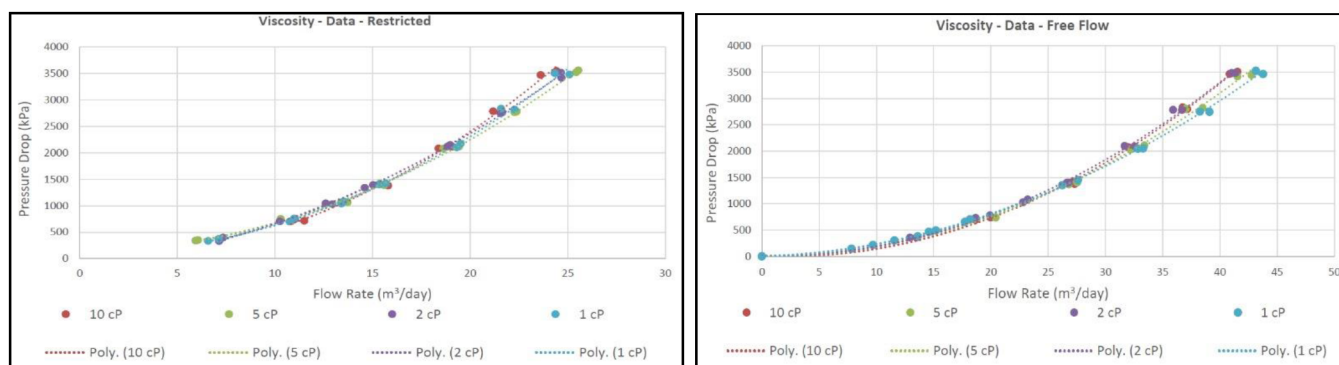


Figure 6—Novel ICD Performance Curve in Restricted and Free Flowing Directions

When flowing from left to right, the fluid paths separate and come back together as the fluid progresses through the device. The splitting and merging of these streams cause higher pressure losses in the device.

The flow from left to right is defined as the "restricted flow direction" and assigned for ICD feature in the production mode. In contrary, when the fluid is flowing in the opposite direction, the fluid path is down the middle of the profile, and the fluid is not splitting and merging where the flow experiences less restrictions (pressure drop). This direction is defined as the "free-flowing direction" allowing higher injection rate (more than 1.5 times) at same pressure drop across the cassette, therefore it is referred as injection mode Fig. (7).

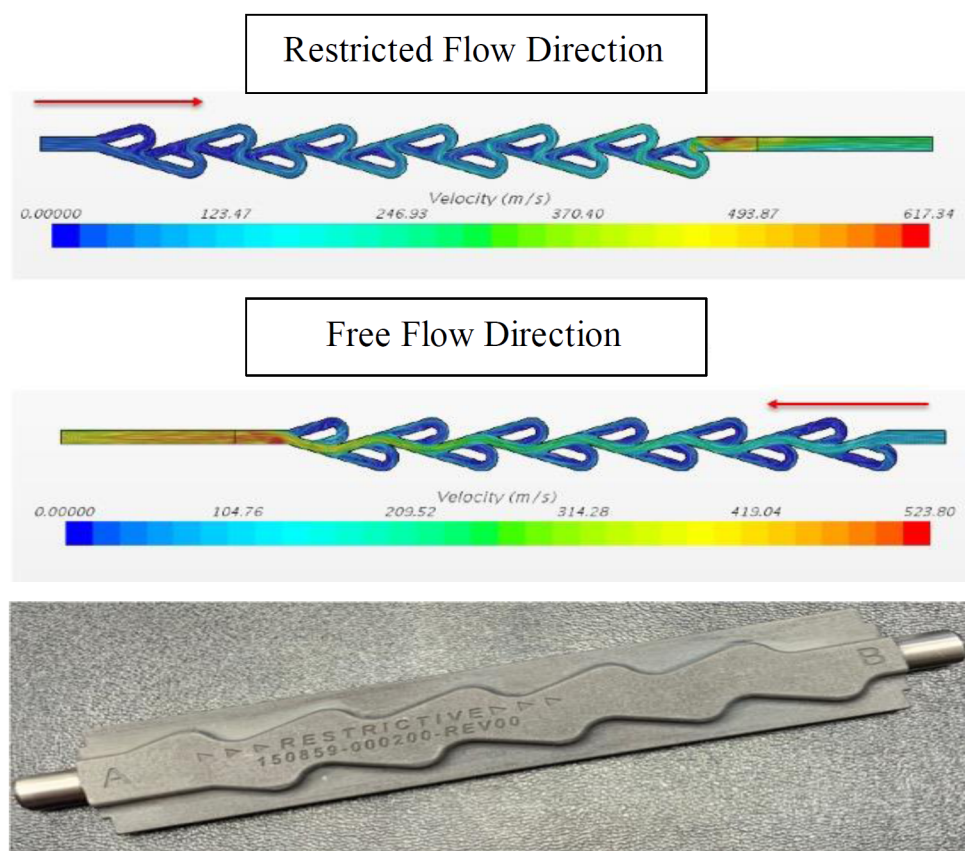


Figure 7—Novel ICD cassette restricted and free flowing directions

The Novel ICD cassette has roughly four times cross section flow area compared with equivalent nozzle-based ICD's, which leads to lower fluids velocities through the device, reducing erosion and making the design less susceptible to plugging. In addition, the gradual pressure changes through potentially reduce scaling tendency in the device, minimizing the number of cleanout intervention operations required throughout the life of the well. The system is applicable for both producer / injector wells and may allow for oil producer wells to be converted to water injectors in future while preserving Outflow Control Devices (OCD) feature. For example, a TOC03 cassette with 2.5mm equivalent nozzle performance (pressure vs flow rate performance) has a flow area equal 0.030 sq.in. A 2.5mm flow control nozzle has a circular flow area equal to 0.0076 sq.in. The flow control cassette then provides 3.95 times (~4 times).

Case Study: Novel ICD installation in Canadian Heavy Oil Wells

Background. An operator in west side of Canada heavy oil well that required a balanced Injection rate utilizing a flow control device as waterflooding solution to maintain reservoir pressure and increase recovery factor. The 4-1/2" perforated liner horizontal well was designed for injection rate around 1,000 bpd to displace 400 cP heavy oil in around 800 mD of reservoir permeability. Base case simulates perforated liner did not include a flow control device for variable reservoir permeability to the simulation with the ranges of 500, 800, and 1,500 mD to reflect the heterogeneity of the reservoir. The flux profile has then shown an

uneven injection profile with high possibility of early water breakthrough. The outflux profile for perforated liner base case reflect the same thump of permeability contrast Fig. (8).

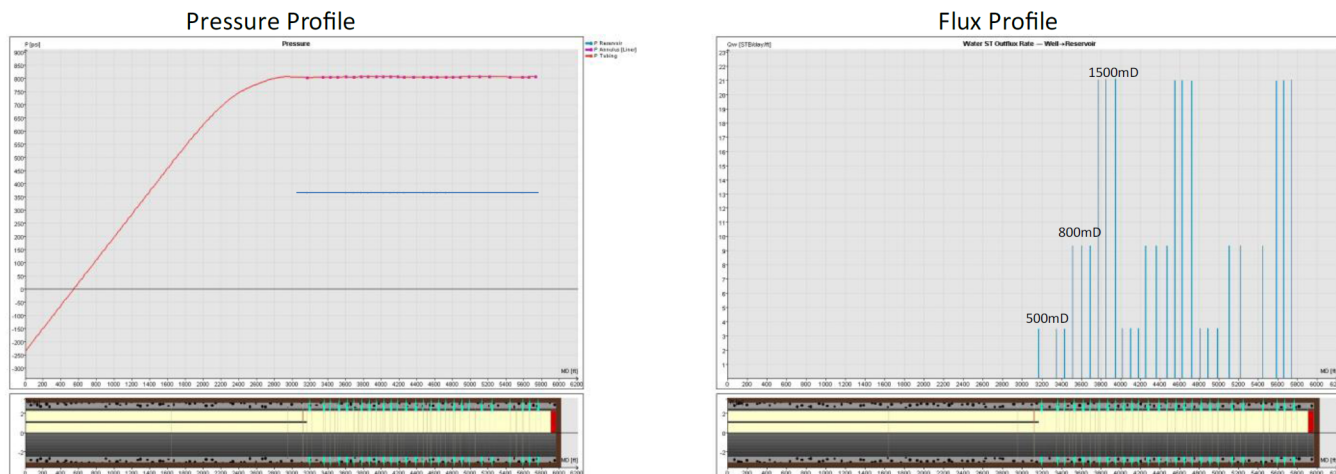


Figure 8—Base Case Simulation with No FCD, Perforated Liner Only, with Varying Permeability

Challenge. The challenge was to design a controlled waterflooding system to balance injection rates across the horizontal well by isolating each stage with packers to ensure balanced injection downhole profile over the varying permeability intervals. Reduce the risk of flow control plugging risk which had been faced with nozzle type.

Design. The design proposed as a nine compartments 2-3/8" system to be installed inside the 4-1/2" perforated liner equal strength cassettes have been used in all compartments with 2 ICD joints for each compartment to provide some injection flexibility depending on results. Variable pressure-drop across downhole completion is generated as shown in Fig. (9) which led to change in outflux profile compared with OH base case; Fig. (10) shows remarkable improvement in the outflux profile.

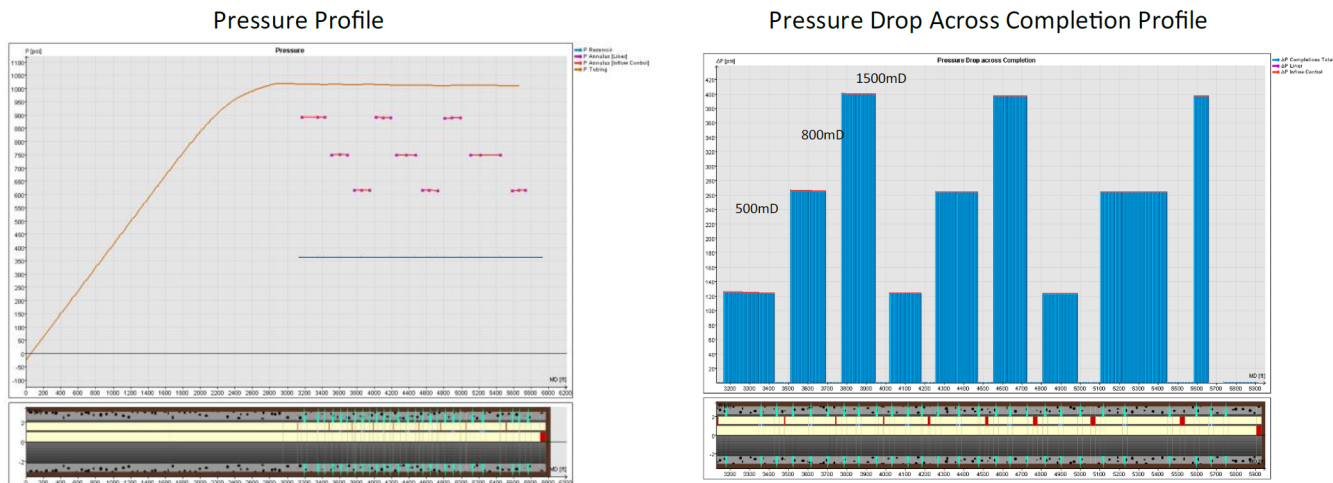


Figure 9—Downhole Pressure Model for the novel ICD

Flux Profile

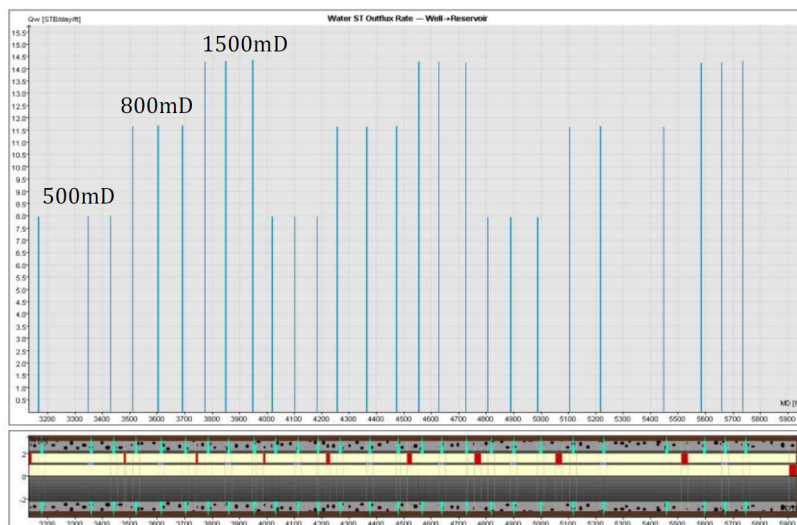


Figure 10—Novel ICD Injection outflux profile with equal strength cassettes

The well was designed for injection rate of 150m³/d (~1,000 bpd), to displace a heavy oil (400 cP), with varying permeability (500, 800, and 1,500 mD) on average across the well.

Design process also includes comparing the base case of perforated liner versus the Novel FCD Injection String. After that, a sensitivity analysis of varying permeability along the lateral section to be completed. To determine appropriate cassette selection to match injection conformance requirements.

The following inputs were used in the NETool Steady State Numerical Simulator for modeling the design:

Table 3—Model Data Input in NETool Simulator

Model Inputs	Comments
Well Construction:	
Deviation Survey	Supplied
Proposed Completion Diameters	2-3/8" Injection String inside 4-1/2" Liner
Well Measured Depth	6,100'
True Vertical Depth	2,400'
Lateral Length	~3,200'
Number of Compartments	9
Perforation Program	3 × 3ft perforation at 60° phasing, 6 SPF, 0.48" entrance hole and 51.8" penetration
Reservoir Properties:	
BHP	360 psi
BHT	77°F
Permeability	500-1,500mD: 800mD field average
Oil API	15
Viscosity	400cP
Oil Saturation	~70%
Operations:	
Max Injection Pressure	~800 psi (6000 Kpa)
Designed Injection Rate	1,000 BPD (Potentially to 2,000 BPD)

Installation

The downhole completion string was installed as planned, ensuring the integrity and readiness of the system for the stimulation operation. A ball was dropped to the shut-off valve to isolate the liner, allowing controlled pressure application. This pressure was then used to set the liner hanger and activate the isolation packers, securing the downhole components in place. Coiled tubing was deployed and equipped with a hydraulic shifting tool to sequentially open the sleeves Fig. (11). Injectivity tests were conducted after each sleeve was opened to evaluate formation response and injection capability. During these tests, a constant pressure was applied, and the corresponding injection rate into the well was monitored using the pump truck.



Figure 11—The 2-3/8" Injection novel ICD and Packers (Left) and Coiled Tubing Operation on Site (Right)

Evaluation and Results

Initial Pressure Response. During initial startup, injection began under vacuum conditions at lower rates, indicating minimal initial resistance. However, a rapid early pressure response was observed, which raised concerns about potential formation restrictions or completion-related issues. As injection rates were increased to reach the target of 1,000 barrels per day (BPD), surface pressures rose sharply, reaching up to 800 psi, the maximum allowable line pressure. This was unexpected, as near-zero injection pressure was anticipated at the target rate based on pre-job modeling and reservoir expectations. The discrepancy between expected and observed behavior suggests the need for further investigation, potentially through a Production Logging Tool (PLT) survey to assess downhole flow profiles and identify contributing intervals.

PLT. An injection log was conducted partway throughout the year to evaluate the zonal contributions of each compartment and verify that all zones were effectively accepting fluid. The primary objective was to confirm that distribution performed as designed and that no compartments were plugged or inactive over time. A comparison between initial injection rates and PLT spinner log results is presented in Fig. (12), highlighting any shifts in flow distribution or performance issues. PLT confirms all ICD joints / compartments contribute to well injection and actual water outflux profile is better than design one. As shown in the graph the ideal horizontal injection line for each compartment should be slightly higher than 11% for each compartment if we assume we can equal injection rate for each compartment. The contrast between PLT results and average line is much less compared with simulation results contrast with average line. Both simulation and PLT results is balanced compared with perforated liner outflux shown in Fig. (8).

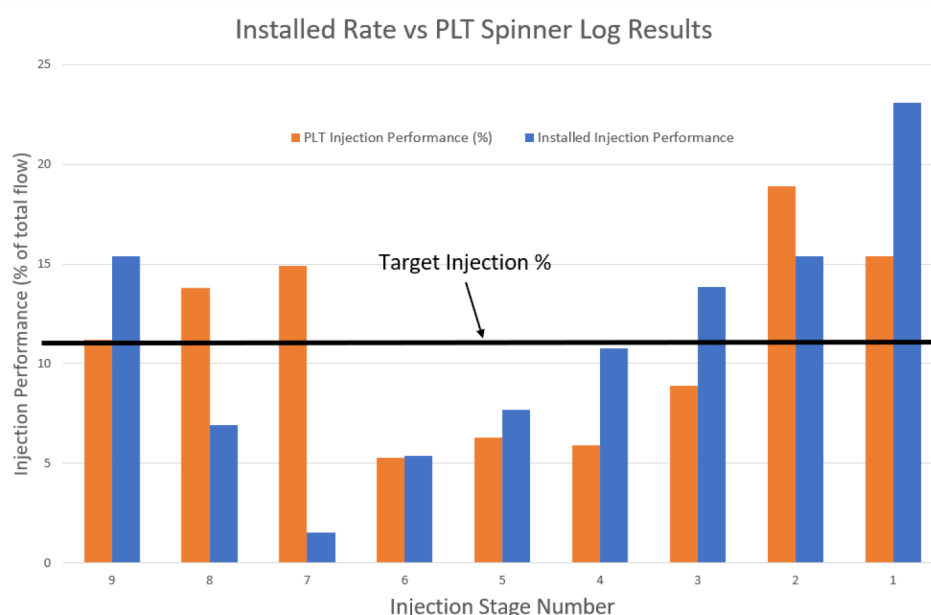


Figure 12—Compartments contribution percentage in injection profile for PLT and simulation Model

It became clear that something was incorrect in either the well condition or the design model. The intent was to maintain low injection pressures to allow for higher future injection rates, but the observed performance contradicted this expectation. A systematic cause analysis was conducted to identify the source of the pressure discrepancy as summarized in [table 4](#). The table below summarizes the potential causes considered and the corresponding findings:

Table 4—High Injection pressure possible cause analysis

Potential Causes	Investigation / Action Taken	Outcome
Several sleeves not open	Injection log confirmed sleeves status	Sleeve confirmed open
Scale formed / plugging of cassettes	Acid soak treatment	No change in injection pressure
Plugged perforations	Acid soak treatment	No improvement observed
Lower permeability than planned	Reassess permeability range	Model updated to reflect realistic lower-end values
Viscosity of the oil	Original assumption revised from 400 to 1,000 cP	Model updated accordingly
Oil saturation	Field Data confirmed oil saturation ranged from 70% to 85%	Added to updated model

Based on above mentioned factors mentioned in [Table-4](#), additional design sensitivity had been done for variable oil viscosity from 400 cP to 1000 cP and oil saturation ranged from 70% to 85%. Sensitivity results shows that saturation of the 1000 cP oil plays a large role on the injection pressure, as shown in [Fig. \(13&14\)](#) a 1000 cP oil saturation 85% has 1000 psi higher injection pressure compared with 1000 cP oil saturation 70%. Variable oil saturation scenarios should be considered in future wells design optimization.

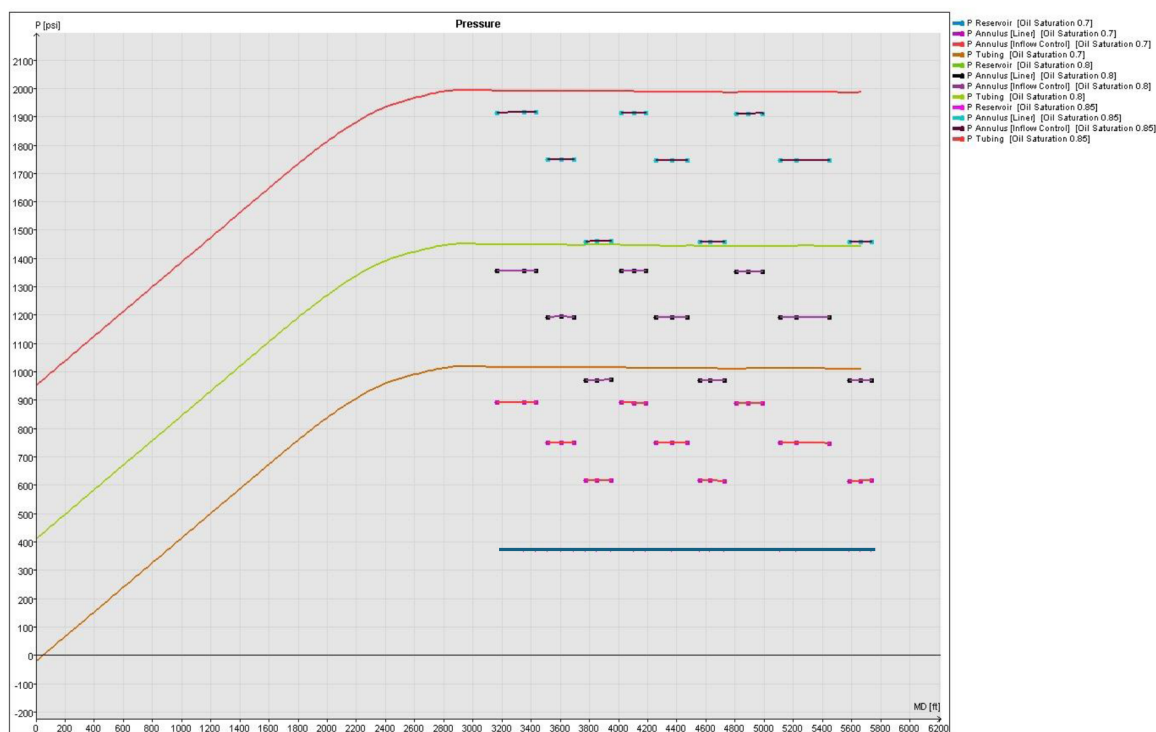


Figure 13—Downhole pressure model for 1000 cP oil different oil Saturation

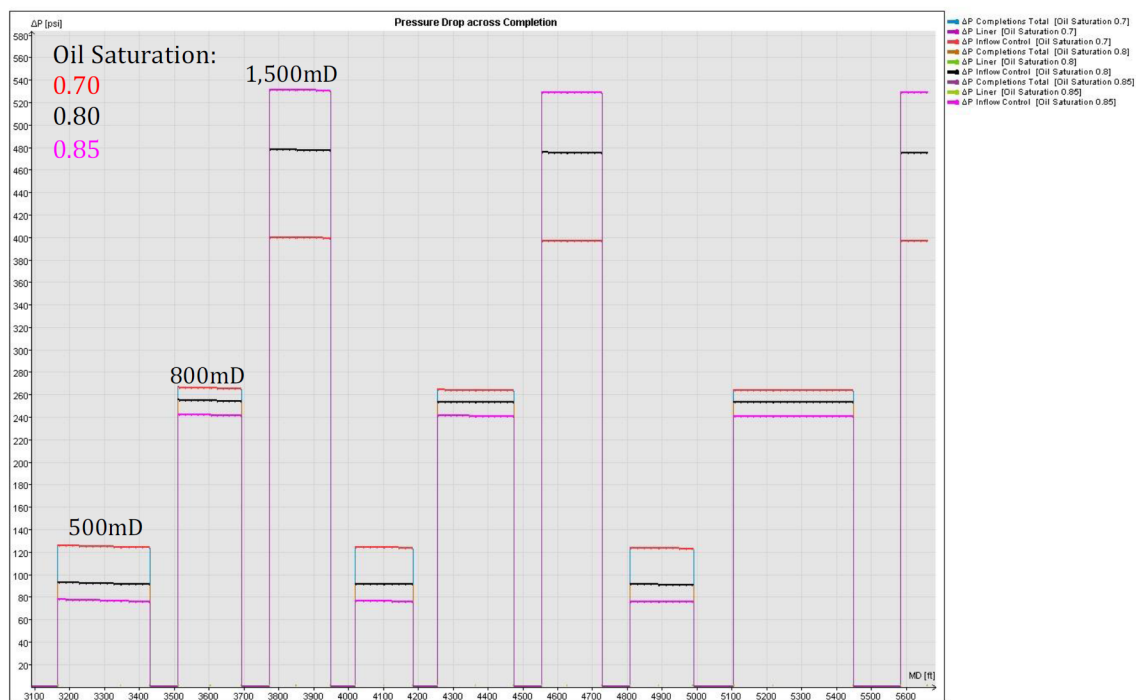


Figure 14—Pressure drop across downhole completion for 1000 cP oil at different oil saturation

Lessons Learned

Designing a Flow Control Device (FCD) injection string for a heavy oil field characterized by high and variable permeability presents significant challenges. To achieve balanced injection, equal choking strength cassettes were utilized, resulting in a relatively uniform distribution across the nine injection compartments. The use of sleeved FCDs provided added flexibility, allowing for the future shutoff of intervals that may

prematurely break through to a nearby producer. Injectivity tests conducted after opening each sleeve served as a valuable diagnostic, confirming sleeve functionality in cases where there was uncertainty due to potential plugging or incomplete opening. However, unexpected field conditions; specifically, a higher-than-anticipated oil viscosity (400 cP observed versus 1,000 cP expected) and varying oil saturations led to inaccuracies in pressure prediction and distribution modeling. Additional design sensitivity scenarios should be run to cover reservoir uncertainties making sure well will deliver the required injection rate within acceptable operating conditions. Extra ICD joint may be added for each compartment – run in close position - which may add additional fixability and room for future design optimization.

Way Forward for Future Wells

Building on previous lessons learned, a new design philosophy has been introduced to improve injection performance and reliability. This approach emphasizes the use of wellsite geologist reports, when available, to better assess reservoir quality prior to finalizing the completion strategy. Additionally, the well is produced ahead of injection operations to create voidage in the near-wellbore region, enhancing injectivity and fluid distribution. Oil samples are collected to accurately determine fluid properties, ensuring that design parameters reflect actual reservoir conditions. An injection log is also recommended prior to completion string installation to confirm injectivity into each perforated interval.

The overall design process is guided by the methodology outlined in SPE 189824, where a pseudo-permeability profile is developed to properly size the FCD cassettes. To manage heterogeneity, especially in zones with higher permeability and lower oil saturation. Variable strength cassettes may be considered to balance more the outflux and achieve required injection rate across the well based on best practices mentioned in (Abd Elfattah, 2013) which discuss in details advantage / limitations of each design optimization method which summarized in tables 5 and 6.

Table 5—Fixed strength FCD design optimization

Advantages	Limitations
Better water breakthrough control even from tight sections.	High pressure drop across ICD with lower flowing bottomhole pressure
Unpredictable open hole reservoir characteristics can be addressed - Uncertainties	Applies unrequired pressure drop across tight zones
Can handle unexpected operational hazards (Stuck – completion cannot reach setting depth)	Have to alter length of compartment for managing pressure drops

Table 6—Variable strength FCD design optimization

Advantages	Limitations
Lower pressure drop across ICD and higher flowing bottomhole pressure.	Uncertainty in permeability profile
Better influx / outflux balance profile.	Unexpected operational hazards like completion getting stuck, unable to reach TD, shift in depths.
Stimulate more the tight reservoir sections by maximizing drawdown against it.	Water encroachment from unpredictable zones.
Allows to put fractures section on production with restricted rate in carbonate reservoir.	Requires visible heterogeneity of differences in permeability across horizontal sections in order to isolate them as separate compartments.

Conclusion

After more than a year the Novel ICD completion with higher cross section flow area does not have any plugging issues compared with offset wells performance. Although the well was completed and activated as per design, the pressure behavior told a different story. What was expected to be a low-pressure injection

system quickly maxed out surface pressure limits, signaling a clear gap between modeled assumptions and actual field response. Initial troubleshooting pointed to mechanical causes, but the injection log confirmed sleeve functionality. Acid soaking ruled out plugging or scale as the issue. In the end, the root cause came down to fluid and reservoir properties; specifically, higher oil viscosity and variable saturation that weren't properly reflected in the original model. This reinforced a key learning: field data must guide design, not the other way around. Moving forward, greater emphasis will be placed on pre-install diagnostics, adaptive design choices like sleeved FCDs, and mid-cycle validation to ensure long-term injectivity and system performance. This well served as a reminder that in heavy oil applications, flexibility, data-driven decisions, and real-time adjustments are essential.

Acknowledgment

The authors gratefully acknowledge the outstanding teamwork and collaboration from engineering teams to accomplish the success of this research and would like to thank PRAIRIE THUNDER RESOURCES LTD and Packers Plus for allowing to publish the case study.

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